Differentiate Between ls, ls -l, and ls -a

Command	Description
ls	Lists files and directories in the current directory
ls -l	Lists files with detailed information (permissions, owner, group, size, date)
ls -a	Lists all files including hidden files (files starting with .)

Use Cases of Linux Commands

a) pwd

- **Displays the present working directory**
- **Used to know the exact location in the file system**

b) whoami

- **Displays the currently logged-in user**
- **Useful for checking user identity, especially after switching users**

c) id

- **Displays user ID (UID), group ID (GID), and group memberships**
- **Used to verify user permissions and access levels**

Steps to Change File Ownership

1. **Check current ownership:**
   ls -l filename
2. **Change file owner:**
   chown newuser filename
3. **Change owner and group:**
   chown newuser:newgroup filename
4. **Verify ownership:**
   ls -l filename

Why Do We Need File Permissions?

File permissions are required to:

- **Protect sensitive files and data**
- **Control who can read, write, or execute a file**
- **Prevent unauthorized access or accidental deletion**
- **Maintain system security and stability**

What Does chmod 777 Do?

The command chmod 777 filename gives:

- **Read, write, and execute permissions to owner, group, and others**

This permission is not recommended for production systems as it allows full access to everyone and can cause security risks.

Summary:
Linux commands help manage files and users, while ownership and permissions ensure system security and controlled access.